UNIT 3 · TOPIC 13

Standards and Standardization

Syllabus topic: Standards and their formulation: organizations, standardization of farm equipment, measurement of specifications

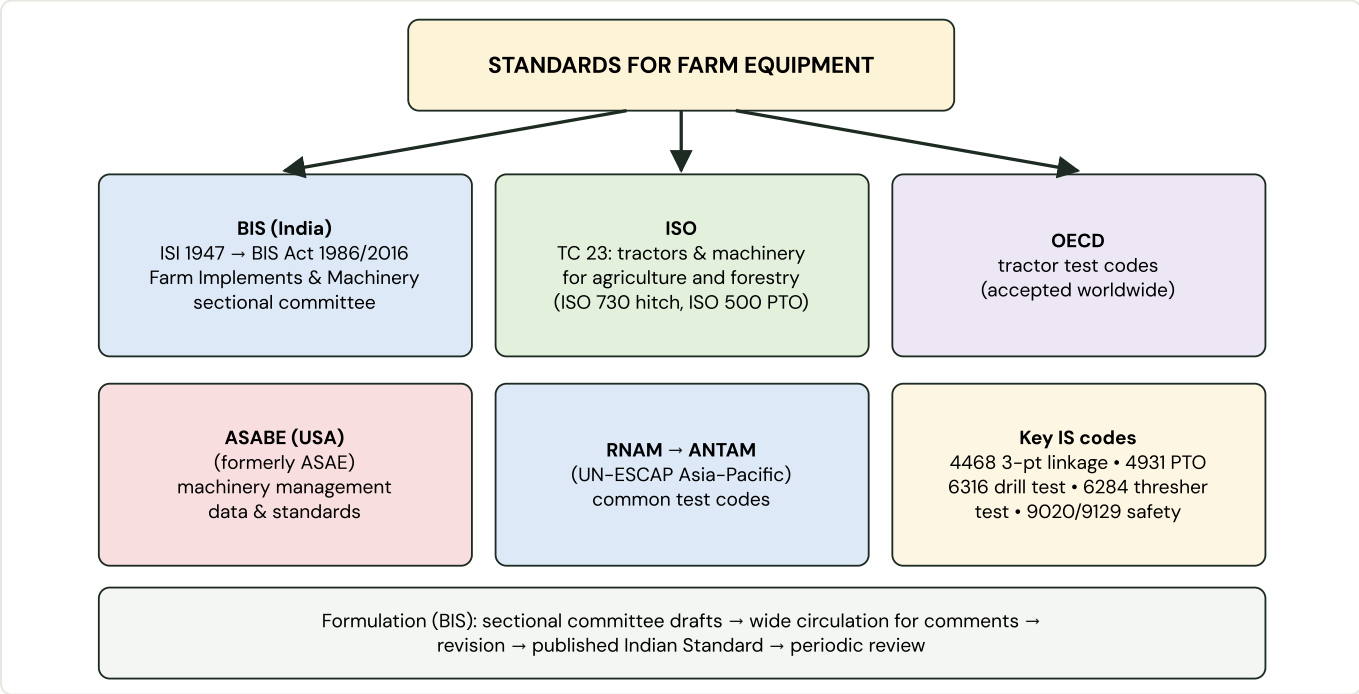
Why standards are needed

- **Interchangeability** of parts — hitch pins and PTO shafts fit any make.
- Assured minimum quality and safety.
- Fair trade and procurement — subsidy eligibility requires conformity/test reports.
- Common test methods, so results are comparable.
- Faster design (standard elements — Unit 2, Topic 1).

How a standard is formulated (BIS route)

1. A need is identified.
2. A sectional committee (manufacturers + users + testing institutes + government) prepares a draft.
3. The draft is circulated widely for public comment.
4. The committee finalises it.
5. It is published as an IS, then periodically reviewed/revised.

In India, the ISI was established in **1947** and became the **Bureau of Indian Standards** under the BIS Act **1986** (updated **2016**); conforming products carry the ISI mark.



Standards for farm equipment — organisations and key codes (redrawn from the source figure)

Organisations and their roles

Organisation	Role
BIS	National standards; ISI mark
ISO	TC 23 covers tractors and agricultural/forestry machinery — ISO 730 (three-point-linkage categories), ISO 500 (PTO types/speeds), ISO 11783 (ISOBUS)
OECD	Internationally accepted tractor performance & safety test codes (Unit 4)
ASABE (ex-ASAE, USA)	Machinery management data (D497 draft equation) and design standards
RNAM → ANTAM (UN-ESCAP)	Harmonised Asia-Pacific test codes
IEC; SAE/DIN	Electricals; sectoral standards

Measurement of specifications

Verifying a machine against its declared specification sheet is **always the FIRST section of any FMTTI test report** (Topic 2a). It covers:

- Overall dimensions and mass; wheel track/clearance.
- Hitch category (Cat I/II three-point linkage per **IS 4468/ISO 730**).
- PTO speed/spline (**540 rpm, 6-spline 35 mm** per **IS 4931/ISO 500**).
- Rated capacities; materials/hardness of critical parts; safety guards and markings.

Practice MCQs

Q172. The ISO technical committee for agricultural tractors and machinery is:

(a) TC 23 (b) TC 176
(c) TC 204 (d) TC 34

Q173. Three-point-linkage hitch categories are defined by:

(a) ISO 500 (b) ISO 730 / IS 4468
(c) ISO 9001 (d) IS 6284

Q174. The Indian Standards Institution (now BIS) was established in:

(a) 1935 (b) 1947
(c) 1965 (d) 1986

Q175. The FIRST section of an FMTTI machine test report is:

(a) Fuel results (b) Specification verification of the machine
(c) Price list (d) Warranty

Answers

Q172 — (a) TC 23

Q173 — (b) ISO 730 / IS 4468

Q174 — (b) 1947 — *ISI 1947; restructured as BIS under the BIS Act 1986 (revised Act 2016).*

Q175 — (b) Specification verification of the machine

Sources: Tractors and their Power Units (Liljedahl et al.); Kepner et al.; BIS publications; ISO/OECD/ASABE documentation; NPTEL lectures on farm power and IC engines.